

Plan operations in multi-step problems

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: A shop receives 48 boxes of 125 books and sells 3,760 books. How many remain? Copy or complete the first step.

2. A club collects £18 from each of 236 members, then spends £1,875. How much remains?

3. Write an operation plan for sharing 9,450 items into 35 equal groups and adding 12 spare items to each group.

4. Miro says: Miro begins calculating every number in the order it appears. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: A shop receives 48 boxes of 125 books and sells 3,760 books. How many remain? Copy or complete the first step.
Answer 48 x 125, then subtract 3,760. The answer is 2,240.
2. A club collects £18 from each of 236 members, then spends £1,875. How much remains?
Answer £2,373.
3. Write an operation plan for sharing 9,450 items into 35 equal groups and adding 12 spare items to each group.
Answer Divide 9,450 by 35, then add 12. Each group has 282 items.
4. Miro says: Miro begins calculating every number in the order it appears. Is this correct? Explain.
Answer Plan the relationships first. The order of numbers in the text may not match the calculation order.

Plan operations in multi-step problems

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. A shop receives 48 boxes of 125 books and sells 3,760 books. How many remain?

6. Check this result using an inverse, estimate or known fact: Write an operation plan for sharing 9,450 items into 35 equal groups and adding 12 spare items to each group.

2. A club collects £18 from each of 236 members, then spends £1,875. How much remains?

7. Write a two-step problem represented by $(64 \times 28) - 950$.

3. Write an operation plan for sharing 9,450 items into 35 equal groups and adding 12 spare items to each group.

8. Identify and correct this misconception: Miro begins calculating every number in the order it appears.

4. Solve this independently and show every stage: A shop receives 48 boxes of 125 books and sells 3,760 books. How many remain?

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: A club collects £18 from each of 236 members, then spends £1,875. How much remains?

10. Write and solve a similar question to: Write an operation plan for sharing 9,450 items into 35 equal groups and adding 12 spare items to each group.

Teacher answers and checking prompts

1. A shop receives 48 boxes of 125 books and sells 3,760 books. How many remain?
Answer 48×125 , then subtract 3,760. The answer is 2,240.
2. A club collects £18 from each of 236 members, then spends £1,875. How much remains?
Answer £2,373.
3. Write an operation plan for sharing 9,450 items into 35 equal groups and adding 12 spare items to each group.
Answer Divide 9,450 by 35, then add 12. Each group has 282 items.
4. Solve this independently and show every stage: A shop receives 48 boxes of 125 books and sells 3,760 books. How many remain?
Answer 48×125 , then subtract 3,760. The answer is 2,240.
5. Use a second method or representation for: A club collects £18 from each of 236 members, then spends £1,875. How much remains?
Answer Expected result: £2,373. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Write an operation plan for sharing 9,450 items into 35 equal groups and adding 12 spare items to each group.
Answer Expected result: Divide 9,450 by 35, then add 12. Each group has 282 items. A valid check must be shown.
7. Write a two-step problem represented by $(64 \times 28) - 950$.
Answer Answers vary. The final answer to the expression is 842.
8. Identify and correct this misconception: Miro begins calculating every number in the order it appears.
Answer Plan the relationships first. The order of numbers in the text may not match the calculation order.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Write an operation plan for sharing 9,450 items into 35 equal groups and adding 12 spare items to each group.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Plan operations in multi-step problems

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Write a two-step problem represented by $(64 \times 28) - 950$.

6. Create a new example that tests the same idea as: Write an operation plan for sharing 9,450 items into 35 equal groups and adding 12 spare items to each group.

2. Prove the answer to this question, rather than only calculating it: A shop receives 48 boxes of 125 books and sells 3,760 books. How many remain?

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: A club collects £18 from each of 236 members, then spends £1,875. How much remains?

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: Divide 9,450 by 35, then add 12. Each group has 282 items.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro begins calculating every number in the order it appears.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Write a two-step problem represented by $(64 \times 28) - 950$.
Answer Answers vary. The final answer to the expression is 842.
2. Prove the answer to this question, rather than only calculating it: A shop receives 48 boxes of 125 books and sells 3,760 books. How many remain?
Answer Expected result: 48×125 , then subtract 3,760. The answer is 2,240. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: A club collects £18 from each of 236 members, then spends £1,875. How much remains?
Answer Expected result: £2,373. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: Divide 9,450 by 35, then add 12. Each group has 282 items.
Answer One valid reconstruction is: Write an operation plan for sharing 9,450 items into 35 equal groups and adding 12 spare items to each group. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro begins calculating every number in the order it appears.
Answer Plan the relationships first. The order of numbers in the text may not match the calculation order.
6. Create a new example that tests the same idea as: Write an operation plan for sharing 9,450 items into 35 equal groups and adding 12 spare items to each group.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.